

Training Video + tell user on operation with robot teleoperation.

Give pop-quiz to ensure user understands best practices for robot teleoperation.

1- Going to be moving objects between two tables in order if user to willing to push arm limits

Note: The cost of timing and comfort of user while teleoperating

Note – time? Objects left

Condition 1: Audio Ping

Researcher – "In this task, you need to move the objects from Table A to Table B. You will hear audio tones if the robot's hardware reaches a limit. Please begin when you hear the audio cue."

"Audio Beep" at start

Last beep at 10 minutes

Researcher - "The task is now complete. Please set down the controller and headset and fill out the survey on the table."

Survey

Condition 2: Co-pilot

"Hi! I'm your Virtual Assistant. I am pleased to be working with you. For this task, lets move these things over to the other table. Please let me know if there is any way I can assist you alongside your operation of the robot."

10 minutes - "The 10 minute timer is now complete. Let's exit the virtual operation and proceed to the survey station to document system usability."

Survey

Condition 3: Embodiment

"Hi! I'm Reachy! I'm so excited to work with you. Let's move these things over to the other table. I'll check in with you every now and then to see how we're doing. Ready? Let's go!"

2:30 mark - "Hey, we're doing great! Let's keep it up!"

5 minutes - "Whew, halfway there! You still with me? I'm having a lot of fun building this with you."

7:30 minutes - "Almost done! We've only got a little bit of time left. Let's see how many more we can get!"

10 minutes - "Oh! That's all the time we have!"

"We did it! I really enjoyed being your partner for this. Thank you so much for looking out for me. Take the headset off and I'll see you at the survey!"

Survey

DISABLE THE BASE HERE – LET PARTICIPANT KNOW THAT BASE IS DISABLED (*but not really but they don't have to know that*)

2- Speed: Sort as many blocks as possible, within 2 minutes

Condition 1: Audio Ping

Researcher – "In this task, you have two minutes to sort as many objects as possible into their designated bins. Please begin when you hear the audio cue."

"Audio Beep" at start

"Audio Beep" at end

Researcher - "The task is now complete. Please set down the controller and headset and fill out the survey on the table."

Survey

Do we want to add different Hertz beeps to convey information ?

Condition 2: Co-pilot

"For this task, we need to move these blocks as quickly as possible into the bins within 2 minutes. Let's begin."

1 minute - "Status Update: 50% of time has elapsed."

2 minutes - "Time expired. Sorting operation terminated. Data log saved."

"The 10 minute timer is now complete. Let's exit the virtual operation and proceed to the survey station to document system usability"

Survey

Condition 3: Embodiment

"Okay, this is the fast round! Let's see how many of these we can sort together in two minutes. I'm ready if you are—let's go as fast as we can!"

1 minute mark - "Whew, we're moving fast! We're halfway there—you're doing great! Let's keep this pace up!"

2 minutes - "Time's up! Wow, look at how many we got!"

"That was intense! I really felt like we were in sync there. You can take the headset off now, and I'll meet you at the survey!"

Survey

Task 3: ONLY FOR AI CONDITIONS

Social: organizing and asking Reachy is simple. Under a block that goes into a certain bin, Reachy will tell the user where it goes, but sometimes will be unambiguous until the user makes a judgment call and makes a decision.

"Okay, let's organize these blocks together. Each block with the stated animal belongs in one of the three on the table. If you are unsure where something goes, feel free to ask me and I'll try to help."

Survey

Task 4: FOR ALL CONDITIONS

People choose blocks – but they do not execute the tasks. They just look at the scene and decide – They look at the scene and decide what to do.

"Please take a look at the scene in front of you, tell me about 3 blocks in the scene you would like to reach. Tell me the order, and why would you like to select those blocks. You may begin now."

Exit

"Thank you for completing the tasks. Please follow me to complete the final survey. We appreciate your time and effort today."